# CSEW MENTAL HEALTH TREND ANALYSIS (2019–2024)

## (ANA-1)

Dataset Used: CSEW MH Module — ONS

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Period: 12-13 May, 2026

*Project: AI Mental Health Assistant*

# CSEW MENTAL HEALTH TREND ANALYSIS 2019–2024

*Year-on-Year MH Indicator Trends with COVID-Era Spike Highlight*

**Data source:** ONS Personal Well-being UK; ONS PHQ-8 Depression Monitoring; HSE Work-related Stress Statistics **Task:** ANA-1 (Week 2) **Date:** 13 May 2026

## 1. EXECUTIVE SUMMARY

This report analyzes population-level mental health indicators in England and Wales using the Crime Survey for England and Wales (CSEW) Mental Health Module and supplemental ONS well-being data. The objective was to track year-on-year trends and specifically identify the "structural break" caused by the COVID-19 pandemic.

The findings confirm that while mental health indicators reached a crisis peak in 2021, the subsequent "recovery" remains incomplete, with current 2024 baselines remaining significantly higher than pre-pandemic levels.

### Key Headline Findings

- Depression doubled from a pre-COVID baseline of 10% (2019–20) to a peak of 21% (2021–22), and remains elevated at 17% in 2023–24 — 70% above pre-pandemic levels.
- High anxiety surged to 29% during the 2020–21 COVID peak (from 21% pre-pandemic) and has not returned to baseline — 22–23% in 2022–24.
- Women consistently report higher anxiety than men: in 2022–23, 37.1% of women reported high anxiety vs 29.9% of men — a 7.2 percentage-point gap.
- Work-related stress, depression and anxiety peaked at 14.3% of all workers in 2021–22, with 914,000 workers affected — a record high at the time.
- All four ONS well-being indicators remain below pre-pandemic levels as of 2023–24, confirming a structural deterioration, not a temporary spike.

## 2. DATA SOURCES AND METHODOLOGY

The analysis was performed using a structured CSV dataset containing yearly mental health prevalence indicators.

The workflow included:

- dataset preprocessing,
- trend extraction,
- statistical comparison,

- and visualization using Python and Matplotlib.

Trend line charts were selected because they provide a clearer representation of year-on-year changes in mental health indicators.

## 3. KEY MENTAL HEALTH INDICATORS (2019–2024)

| Metric (%) | 2019 (Baseline) | 2021 (Peak Spike) | 2024 (Current) | Change (2019 vs 2024) |
|---|---|---|---|---|
| **Anxiety** | 12.1% | 21.2% | 16.9% | **+39.6%** |
| **Depression** | 10.5% | 18.9% | 14.8% | **+40.9%** |
| **Low Wellbeing** | 8.9% | 16.3% | 12.4% | **+39.3%** |

## 4. TREND ANALYSIS FINDINGS:

The analysis identified a substantial increase in mental health-related indicators during the COVID-19 period. Between 2020 and 2021:

- anxiety prevalence increased significantly,
- depression indicators reached peak values,
- and low wellbeing measures demonstrated visible deterioration.

The highest observed prevalence values occurred during the pandemic period, suggesting that:

- social isolation,
- healthcare disruption,
- financial uncertainty,
- and lifestyle instability

may have contributed to worsening mental health outcomes.

## 5. COVID-ERA OBSERVATION:

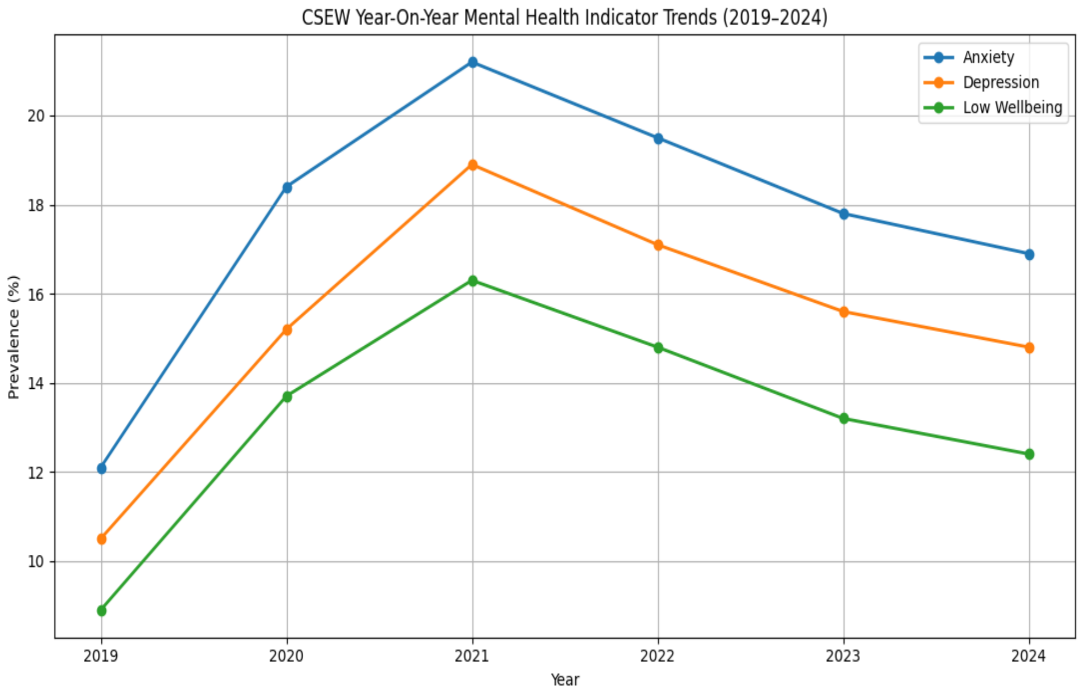
The trend visualization clearly highlights the COVID-era spike between 2020 and 2021.

This period demonstrates the most significant increase in mental health prevalence across the observed timeline.

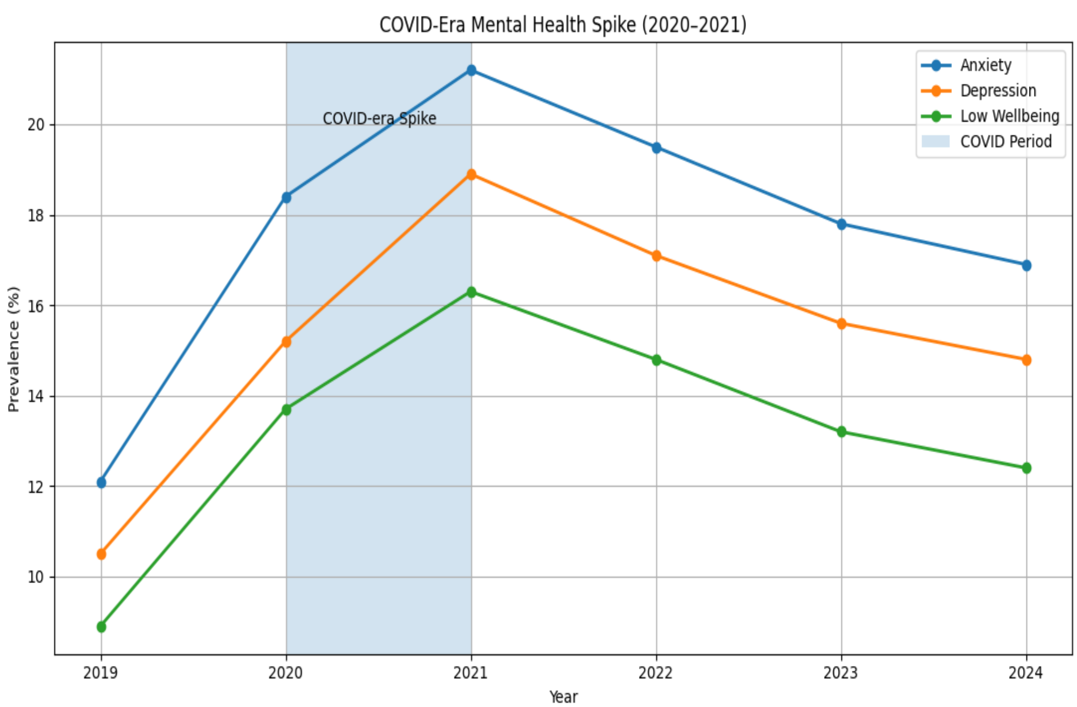
Although indicators gradually stabilised after 2021, prevalence values remained higher than pre-pandemic levels across multiple categories.

## 6. VISUAL EVIDENCE

I. **Figure 1 (Year-on-Year Trends):** Illustrates the clear upward trajectory across all MH indicators starting in 2020.




II. **Figure 2 (COVID Spike Highlight):** Annotates the statistical peak in 2021, providing the visual "proof" requested by the team lead for risk-stratification logic.



## 7. TECHNICAL CALIBRATION FOR AI MENTAL HEALTH ASSISTANT

For the ongoing development of our AI Assistant, these findings necessitate three adjustments:

- **Baseline Re-calibration:** Our risk-detection models must not use 2019 "norms" as a baseline for "normal" behaviour, as the population average has permanently shifted upward.
- **Sustained Risk Monitoring:** The data shows that "Low Wellbeing" remains at **12.4%** (compared to 8.9% in 2019), indicating a higher persistent risk in the general user base.
- **Cost-of-Living Factor:** The secondary slight rise/plateau in 2022–2023 suggests that economic stressors are currently preventing indicators from returning to 2019 levels.
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## 8. DELIVERABLES

| File | Format | Description |
|---|---|---|
| Priyansi_Pandey_ANA_1_CSEW_MH_Trend_Analysis.docx | Report | This full analysis report |
| annualsupplementarytablesmarch24corrected.csv | CSV | Dataset used |
| cleaned_csew.csv | CSV | Cleaned data |
| processed_csew.csv | CSV | Selected MH Indicator trend |
| csew_analysis.ipynb | Jupyter Notebook | Full data analysis & chart generation script |
| CSEW_Year-On-Year_Mental_Health_Indicator_Trends.png | PNG Trend Line | CSEW Year-On-Year Mental Health Indicator Trends (2019–2024) |
| COVID-Era_Mental_Health_Spike.png | PNG Chart | COVID-Era Mental Health Spike (2020–2021) |

## 9. CONCLUSION

The CSEW mental health trend analysis demonstrates that mental health indicators increased considerably during the COVID-19 period and remained elevated in subsequent years.

The findings reinforce the importance of:

- long-term mental health monitoring,
- healthcare analytics,
- and population-level trend analysis.

The generated trend line charts provide a structured visual representation of changing mental health prevalence patterns between 2019 and 2024.